

Junior Makers · Class 12

# Phone-Controlled RGB Mood Lamp

Change the colour of your lamp from a phone!

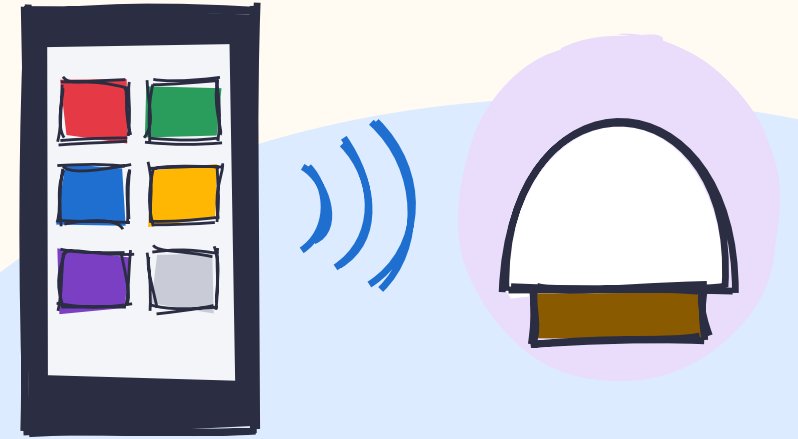
RGB colour mixing

Bluetooth HC-05

MIT App Inventor

Did you know?

Your screen makes every colour you see from just three: red, green and blue.





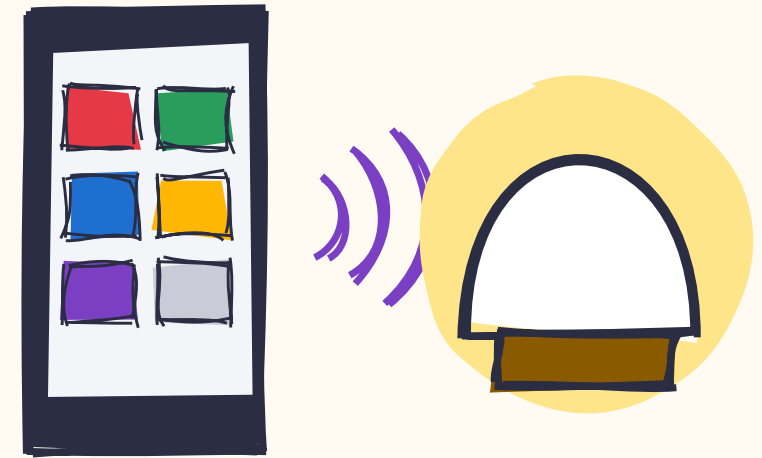
Let's think...

# How does a phone talk to a lamp?



## No wires!

Smart bulbs change colour when you tap a phone. There is **no wire** between them. So how does the message get there?



Magic?

Through the air?

Radio waves!

**Today:** we build a lamp that mixes any colour, and control it from an **Android phone** over Bluetooth.





## Learning aims

# Today I will learn to...



1



Mix colours with an **RGB LED** and PWM

2



Connect a **Bluetooth HC-05** module safely

3



Use an **App Inventor** app to send colour letters

4



Add a **goodnight tune** with the buzzer

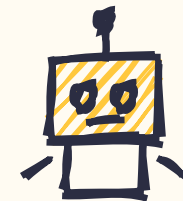
5



Explain how devices send **small messages** wirelessly

### Keywords

RGB · mix · Bluetooth · pair · message  
· app · command



Watch and think

# Video time!



## Colored Light Shadows: Additive Light Mixing

Homemade Science with Bruce Yeany · about 3 minutes ·  
[youtu.be/eKj1EwJ7THU](https://youtu.be/eKj1EwJ7THU)

### Watch for...

- 1 What colours of light does he start with?
- 2 What colour appears where **all three** lights overlap?
- 3 Why are the shadows different colours?

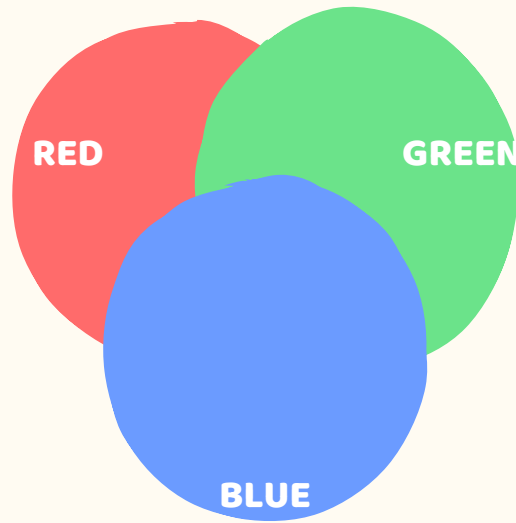


**After the video:** Predict what colour red + green light makes. Test it on your RGB LED!



The science

# Mixing LIGHT is not like mixing paint!



| Light mixed  | Colour you see |
|--------------|----------------|
| red + green  | yellow         |
| red + blue   | magenta        |
| green + blue | cyan           |
| all three    | white          |
| none         | off (black)    |

## Paint vs light

Mix all paints and you get **muddy brown**. Mix all lights and you get **white**!



## Look closely

Use a magnifying glass on a phone screen.  
You will see tiny red, green and blue dots!





Real world

# Bluetooth is all around us



## Smart bulbs

Change colour and brightness from an app



## Earbuds

Music travels from phone to ears by Bluetooth



## Game controllers

Send button presses to the console



## Smartwatches

Show phone messages on your wrist



Name a Bluetooth gadget in your home. What messages does it send?

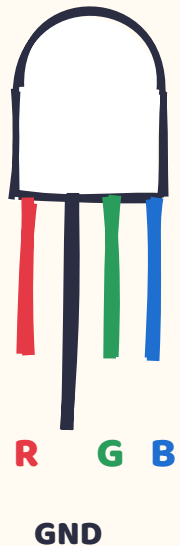
## Short range

Bluetooth works over about 10 metres, inside one room or house.



Meet the light

# The RGB LED: 3 lights in 1



## Inside the bulb

A red, a green and a blue LED share one bulb.  
The **longest leg** is the common ground (GND).

## Brightness 0 to 255

Each colour gets a number using PWM.  
Orange = red 255, green 80, blue 0.

| Leg      | Connect               |
|----------|-----------------------|
| R        | pin 9 + 220 $\Omega$  |
| G        | pin 10 + 220 $\Omega$ |
| B        | pin 6 + 220 $\Omega$  |
| long leg | GND                   |

## 3 resistors

Each colour needs its OWN 220  $\Omega$  resistor, or one colour will hog the power.





Meet the radio

# The HC-05 Bluetooth module



| Pin | Connect                 |
|-----|-------------------------|
| VCC | 5V                      |
| GND | GND                     |
| TXD | pin 2 (Arduino listens) |
| RXD | pin 3 through a divider |

## 3.3 V ears!

The HC-05 **talks** fine to the Arduino, but its **RXD** pin can be damaged by 5 V. Two resistors (**1k** and **2k**) make a **divider** that brings 5 V down to about 3.3 V.

## Android only

iPhones cannot talk to the HC-05. Use an Android phone or tablet.

## Pairing code

When the phone asks: **1234** (sometimes 0000).

## LED blinks

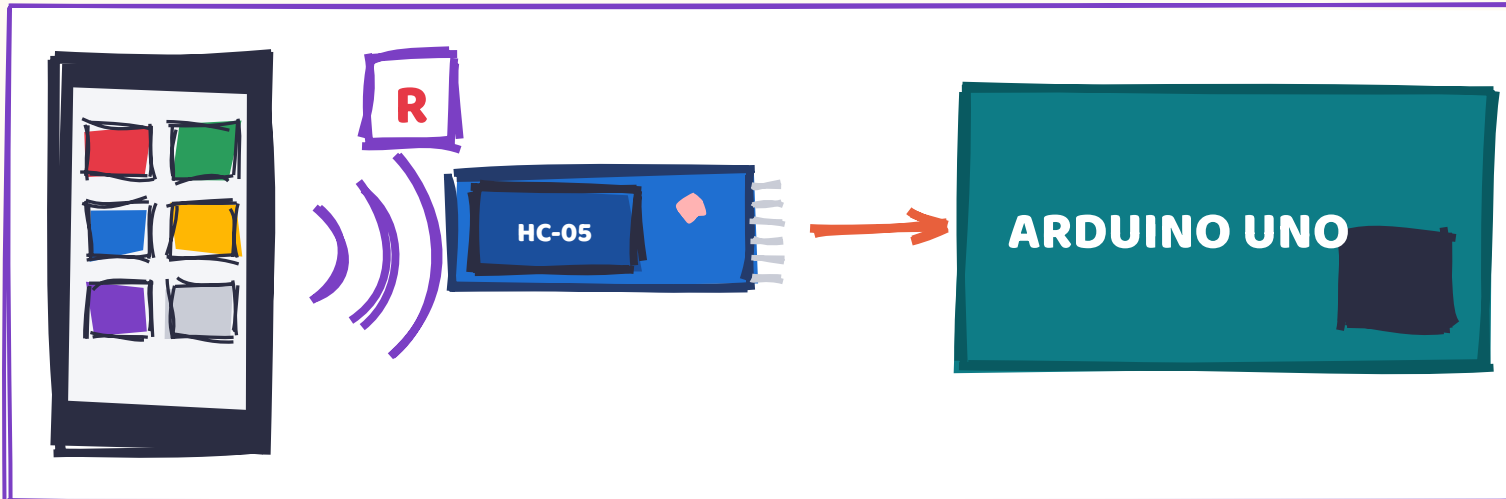
Fast = waiting. Slow double blink = connected!





Big idea

# The app sends ONE letter at a time



## Why letters?

A letter is a **tiny** message: quick to send and easy for the Arduino to check. Real smart bulbs also send short **commands**.



Same as...

Morse code in Little Explorers! A short code that means something bigger.



## Class game

# Be the lamp!

### How to play

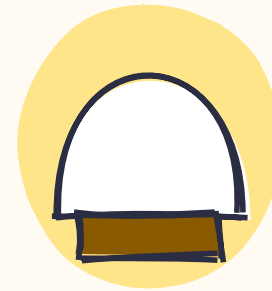
1. One student is the **phone**. Everyone else is a **lamp**.
2. Lamps hold up red, green and blue paper squares.
3. The phone calls a letter: "Y!"
4. Lamps hold up the right colours together (Y = red + green).
5. Wrong colour? Sit down!

### Then discuss

What happens if the phone says a letter the lamp does not know, like "Z"?

### Big idea

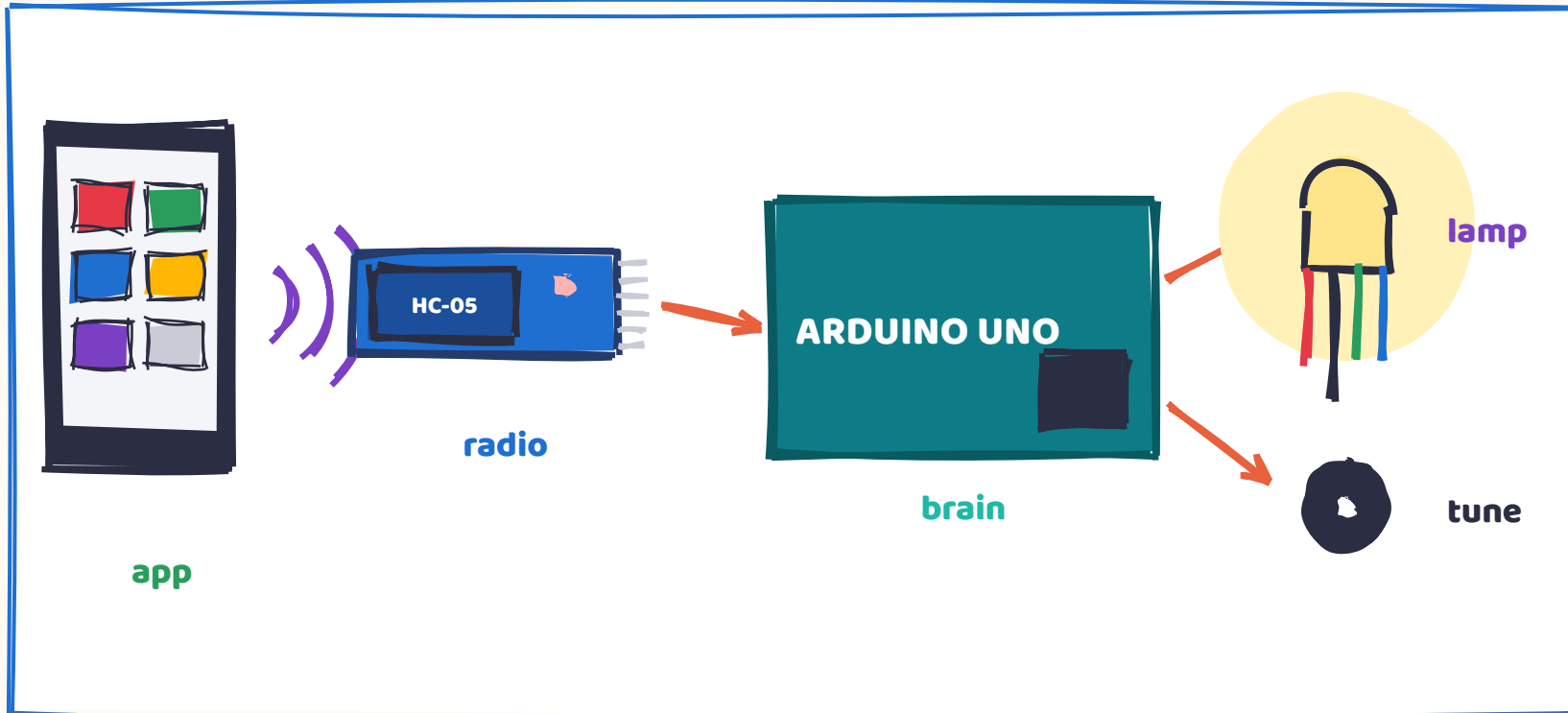
Both sides must agree on the code BEFORE they talk. Engineers call this a protocol.





The plan

# App, radio, brain, lamp



## Two programs!

One runs on the **phone** (App Inventor).

One runs on the **Arduino** (ArduinoBlocks).

They share the letter code.

## Team jobs

Builder · App designer · Tester. Swap next time!





Build it

# Our mood lamp kit (per group)



**Arduino Uno**

The brain



**HC-05 module**

Bluetooth radio



**RGB LED**

With  $3 \times 220 \Omega$



**1k + 2k resistors**

The divider



**Buzzer**

Goodnight tune



**Android phone**

With the app

Also: breadboard, jumper wires, USB cable, laptop, a paper cup or ping-pong ball as a lampshade.

**Lampshade trick**

Put a ping-pong ball over the LED. The colours mix much better!





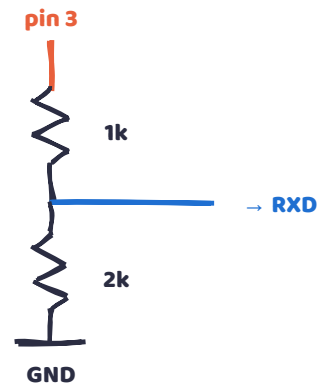
Build it

# Wiring the mood lamp



| Part                   | Connect to                |
|------------------------|---------------------------|
| HC-05 VCC / GND        | 5V / GND                  |
| HC-05 TXD              | pin 2                     |
| Arduino pin 3          | 1k → HC-05 RXD            |
| HC-05 RXD              | also 2k → GND             |
| RGB red / green / blue | pins 9 / 10 / 6 (+ 220 Ω) |
| RGB long leg           | GND                       |
| Buzzer +               | pin 8                     |

## The divider



## No 2k?

Use two 1k resistors in a line.  $1k + 1k = 2k$ !

## Why pins 2 and 3?

Pins 0 and 1 are used by the USB cable to upload code. Keep them free!



Code it

# The lamp in ArduinoBlocks

when Arduino starts

Bluetooth RX pin 2 · TX pin 3 · 9600 baud

forever

if Bluetooth data available

set msg to Bluetooth read

if msg = R → RGB (255, 0, 0)

if msg = G → RGB (0, 255, 0)

if msg = B → RGB (0, 0, 255)

if msg = Y → RGB (255, 180, 0)

if msg = O → RGB (0, 0, 0)

if msg = N → play goodnight tune, then RGB (0, 0, 0)

## Read it like a story

Wait for a letter.

Read it.

Check which letter it is.

Set the three colour numbers.

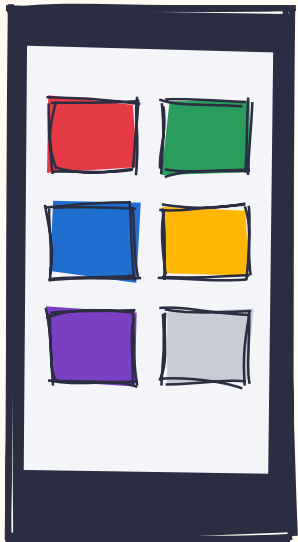
**RGB (r, g, b)**

means: analog write red pin 9, green pin 10, blue pin 6. Make a FUNCTION so you write it once!



The app

# MIT App Inventor



## Designer: what you see

A **ListPicker** button called Connect  
One **Button** per colour  
A hidden **BluetoothClient**

## Ready-made app

The teacher shares a starter **.aia** file. Open it at [ai2.appinventor.mit.edu](https://ai2.appinventor.mit.edu), then test with the **MIT AI2 Companion** app.



when **ListPicker.BeforePicking** → **Elements = BluetoothClient.AddressesAndNames**

when **ListPicker.AfterPicking** → **BluetoothClient.Connect (Selection)**

when **RedButton.Click** → **BluetoothClient.SendText "R"**





Connect it

# Pair the phone with the lamp



- 1 Power** the Arduino. The HC-05 light blinks fast.
- 2 Pair** in the phone's Bluetooth settings: choose HC-05, type 1234.
- 3 Open** your app and press Connect. Choose HC-05 from the list.
- 4 Check** the HC-05 light now blinks slowly: connected!
- 5 Tap** a colour button. The lamp changes!

## App can't find it?

On newer Android phones, allow **Nearby devices** (or Location) permission for the app in Settings.

## Quick test

No app yet? Use the free Serial Bluetooth Terminal app and type R.





Test like an engineer

# Does every colour work?



| Test                   | Expect               | Result      |
|------------------------|----------------------|-------------|
| Tap R                  | red                  | pass / fail |
| Tap G                  | green                | pass / fail |
| Tap B                  | blue                 | pass / fail |
| Tap Y                  | yellow (red + green) | pass / fail |
| Tap N                  | tune, then off       | pass / fail |
| Walk 10 m away and tap | ?                    | pass / fail |

## Debug checklist

Wrong colours? Check which leg goes to which pin.

Nothing happens? Is the TXD wire on pin 2?

Upload fails? Unplug the HC-05 wires from pins 0 and 1.



Swap phones with another team. Can THEIR phone control YOUR lamp? Why could that be a problem?



Level up!

# Mood lamp challenges

## Choose your challenge



**Your colour:** add a new button and letter for your favourite colour (try pink: 255, 40, 120).



**Bedtime mode:** fade slowly from bright to off over 10 seconds.



**Rainbow:** a button that cycles through every colour.



**Slider:** use an App Inventor slider to send brightness numbers.

## Think like an engineer

Anyone nearby with an app could connect to your lamp. How could you stop strangers controlling it?

## Boss level

Change the HC-05 pairing code from 1234 to your own secret code.



Show what you know

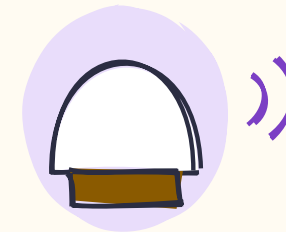
# What did we learn?

- ☒ Mixing red, green and blue light makes any colour
- ☒ Each RGB colour gets a number from 0 to 255
- ☒ Bluetooth sends messages by radio, no wires
- ☒ Phone and lamp agree on a letter code (a protocol)
- ☒ The HC-05 RXD pin needs a divider for 3.3 V



## At home

Find 3 things at home that use Bluetooth. Which one sends the most messages?



**Next class:** a  
Step Counter  
Fitness Tracker!



**Exit ticket:** which two colours of light make yellow?

